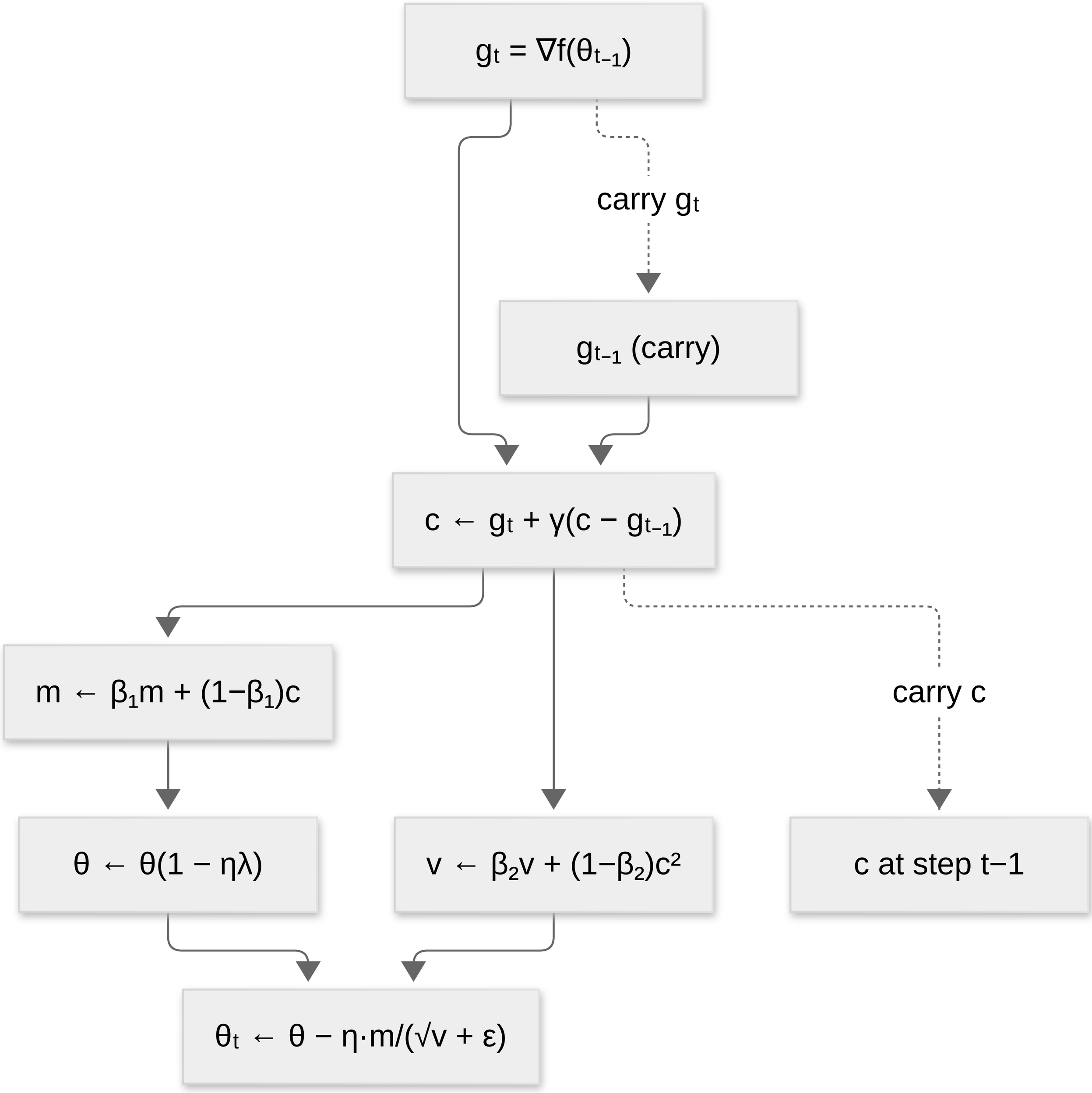

$g_t = \nabla f(\theta_{t-1})$
carry g_t
g_{t-1} (carry)
$c \leftarrow g_t + \gamma(c - g_{t-1})$
$m \leftarrow \beta_1 m + (1-\beta_1)c$
carry c
$\theta \leftarrow \theta(1 - \eta\lambda)$
$v \leftarrow \beta_2 v + (1-\beta_2)c^2$
c at step t−1
$\theta_t \leftarrow \theta - \eta \cdot m/(\sqrt{v} + \varepsilon)$